9th Class Physics – Past Paper 2021

Punjab Board (Rawalpindi Board)

Group-I (Paper Code: 5473)

Objective Type

Marks: 12
Time: 15 Minutes

Multiple Choice Questions (MCQs)

Q.No.	Question	A	B	C	D
1	Density formula	mass × volume	volume/ mass	area/mass	**mass/ volume** ✓
2	Material with large temperature coefficient of linear expansion	**Aluminium** ✓	Steel	Gold	Brass
3	If wall thickness is doubled, thermal conductivity	becomes double	remains same	**becomes half** ✓	becomes 1/4
4	200 μs interval equivalent	0.2 s	0.02 s	**2×10^{-4} s** ✓	2×10^{-4} s
5	Displacement ÷ time =	Speed	Acceleration	**Velocity** ✓	Deceleration
6	Newton's First Law is valid in absence of	Velocity	**Net Force** ✓	Torque	Momentum
7	SI unit of acceleration	m s^{-1}	**m s^{-2}** ✓	m^2 s	m s^2
8	10 N force at 30° – horizontal component	4 N	5 N	7 N	**8.7 N** ✓
9	Torque (τ) equals	F × m	F × R^2	F × τ	**F × L** ✓
10	g on Moon = 1.6 m s^{-2}. Weight of 100 kg body on Moon	100 N	**160 N** ✓	1000 N	1600 N
11	If velocity doubles, kinetic energy	same	doubles	4×	**becomes half** ✓

Q.No.	Question	A	B	C	D
12	Energy of moving body	Nuclear	Chemical	**Kinetic** ✓	Potential

Subjective Type

Marks: 48
Time: 1 Hour 45 Minutes

Section – I

Question 2

Write short answers of any FIVE.

1. Differentiate between scalars and vectors.
2. What is meant by vernier constant?
3. What are prefixes? Give an example.
4. Define speed and velocity.
5. What is the law of inertia?
6. Define acceleration and write its formula.
7. Write two ways to reduce friction.
8. What is the difference between base and derived quantities?

Question 3

Write short answers of any FIVE.

1. Define like and unlike parallel forces.
2. Define geostationary orbit.
3. Define clockwise and anticlockwise moment.
4. Define force of gravitation and give two examples.
5. What is meant by an ideal system?
6. Define torque and write its formula.
7. What is the unit of power? Define its unit.
8. What is meant by satellite and natural satellite?

Question 4

Write short answers of any FIVE.

1. Define temperature and write its SI unit.
2. What is meant by atmospheric pressure?
3. Define density and write its SI unit.
4. What is meant by thermal equilibrium?
5. State Hooke's Law.
6. Why does transfer of heat in fluids take place by convection?
7. Differentiate between conduction and convection.
8. Define specific heat capacity and write its SI unit.

Section – II

Attempt any TWO questions.

Question 5

(a)

With the help of a speed-time graph, prove that:

$V_f = V_i + at$
(4 Marks)

(b)

How much is the force of friction between a wooden block of mass 5 kg on a horizontal marble floor?

Coefficient of friction (μ) = 0.6
(5 Marks)

Question 6

(a)

Define equilibrium and explain the first condition of equilibrium.

(4 Marks)

(b)

Calculate the power of a pump which can lift 200 kg of water through a height of 6 m in 10 seconds.

(5 Marks)

Question 7

(a)

State Pascal's Law and explain the hydraulic press.

(4 Marks)

(b)

A brass rod is 1 m long at 0°C. Find its length at 30°C.

Given:

- Coefficient of linear expansion of brass = $1.9 \times 10^{-5}\ K^{-1}$

(5 Marks)

End of Paper